Recommendation Systems for Personalized Content Discovery

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1. Problem Understanding:

Recommendation systems play a crucial role in helping users discover relevant content on platforms such as Netflix, Amazon, and Spotify. By learning from historical user-item interactions, they can predict user preferences and provide personalized recommendations that improve user engagement and retention.

In this project, we use the Netflix Prize dataset to develop and evaluate a movie recommendation system. The task involves two related but distinct objectives:

- Rating Prediction: Estimating the rating a user would assign to an unseen movie, evaluated using RMSE.
- Recommendation Ranking: Ranking movies so that items a user is likely to enjoy appear at the top of the recommendation list, evaluated using MAP@10. A movie is considered relevant if its actual rating is at least 3.5.

To study different recommendation strategies, two models were implemented and compared: Item-Based Collaborative Filtering (Item-CF) and Singular Value Decomposition (SVD). The models were evaluated on prediction accuracy, recommendation quality, computational efficiency, memory usage, scalability, and interpretability. The objective is not only to predict ratings accurately but also to generate meaningful and personalized recommendations.

2. Dataset Description:

The Netflix Prize dataset is a benchmark dataset widely used in recommendation system research. The complete dataset contains over 100 million ratings from 480,189 users on 17,770 movies, with ratings recorded on a 1–5 scale along with timestamps.

For this project, we used the first rating file (*combined_data_1.txt*) together with the movie metadata file. The selected subset contains:

Property	**Value**
Ratings	24,053,764
Users	470,758
Movies	4,499
Rating Scale	1–5
Date Range	Nov 1999 – Dec 2005

Data Parsing and Preprocessing

The raw Netflix rating files are not stored in a standard tabular format. Each movie appears as a header line followed by its associated user ratings. The data was parsed and converted into a structured table containing four attributes: **MovieID, UserID, Rating, and Date**.

To improve efficiency, numeric columns were downcast to smaller data types and the Date column was converted to a datetime format. Data validation confirmed that there were no missing values and all ratings were within the valid 1–5 range. The processed dataset was stored in Parquet format for faster loading in subsequent stages.

Although only one of the four rating files was used initially, the dataset still contained more than 24 million ratings, making data reduction and filtering necessary before model development.

3. Exploratory Data Analysis:

Smart Subsetting

The original dataset contained over 24 million ratings from 470,758 users and 4,499 movies. To make recommendation modelling computationally feasible, only users with at least 400 ratings and movies with at least 2,000 ratings were retained.

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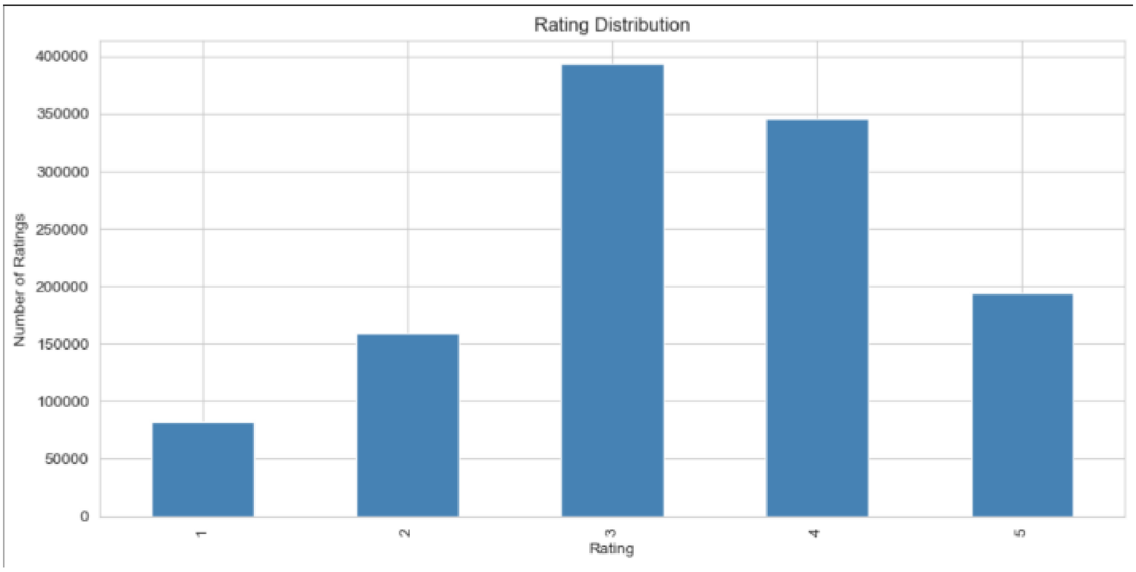
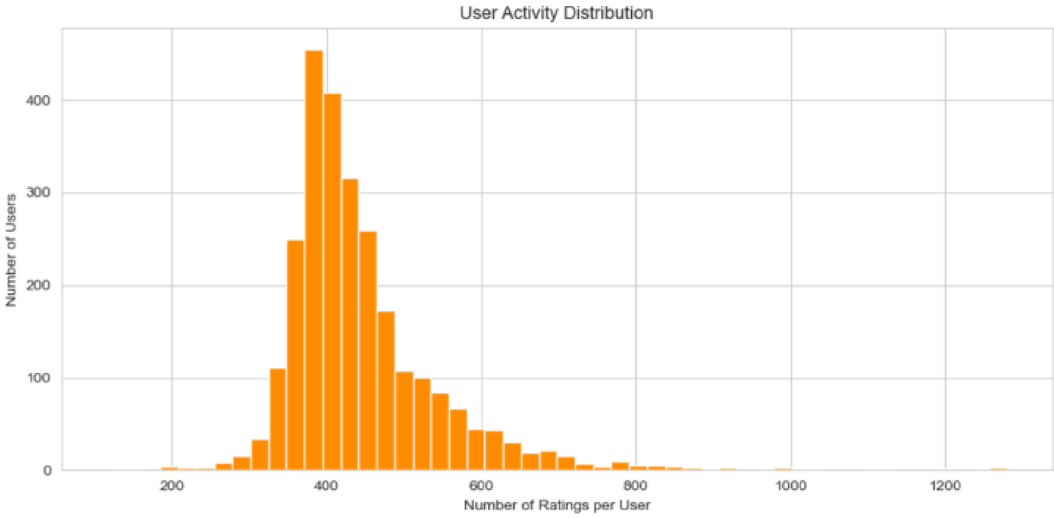
iginal	,053,764	0,758	99
tered	75,260	34	87

This filtering produced a denser and more manageable dataset for collaborative filtering and matrix factorization.

Rating Distribution and User Activity

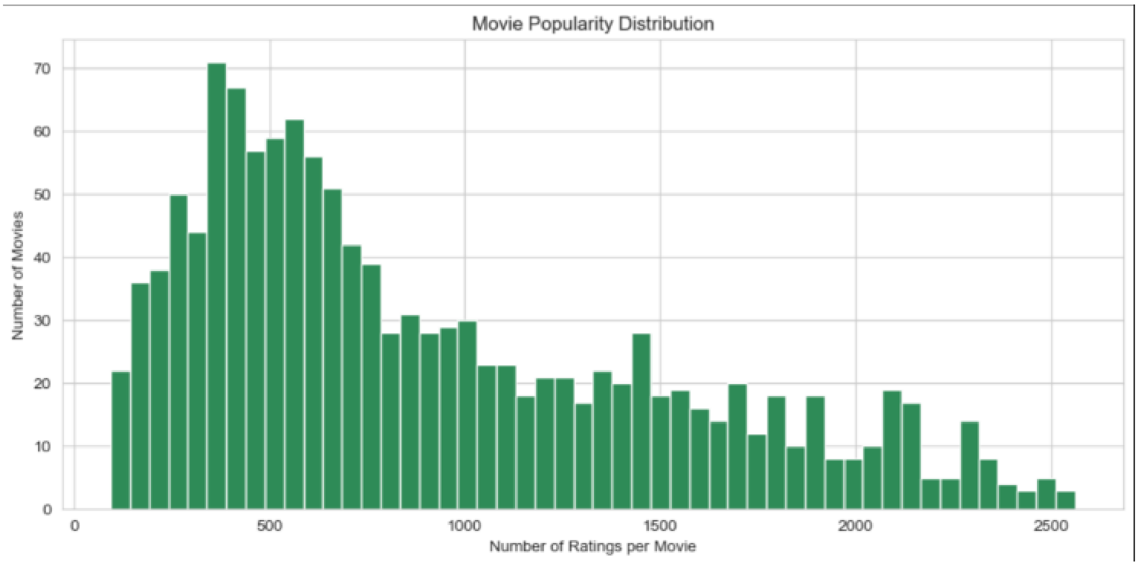
Ratings are concentrated around the middle of the scale, with rating 3 being the most common. The average rating is 3.35 and the median is 3.0. User activity is highly uneven, with a small number of power users contributing substantially more ratings than the average user.

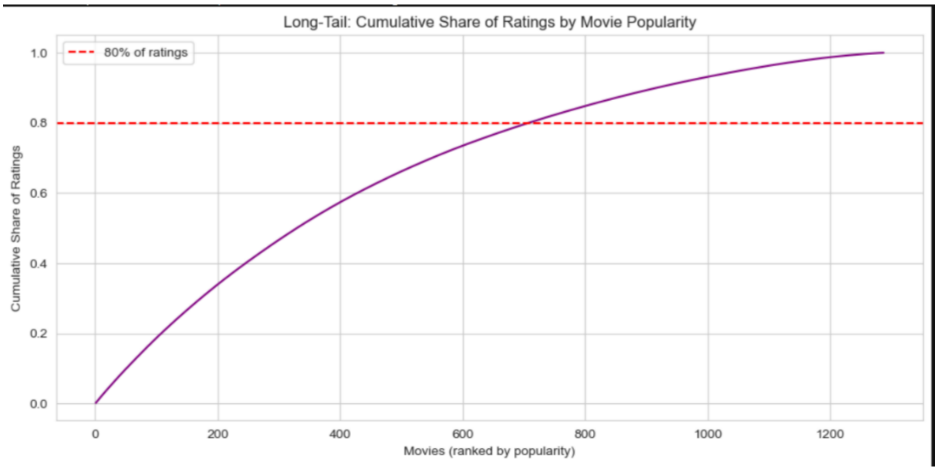




Movie Popularity and Long-Tail Analysis

Movie popularity follows a right-skewed distribution where a small number of blockbuster titles receive a large proportion of ratings. The dataset also exhibits a long-tail pattern, with 708 movies (55% of the catalogue) accounting for 80% of all ratings.





Sparsity and Key Insights
The filtered user-item matrix has a density of 34.7% and a sparsity of 65.3%, making it substantially denser than the original Netflix dataset.
Key observations include:

- User activity is uneven, with a small group of highly active users.
- Ratings are centred around 3, indicating a moderately critical user base.
- Popularity is concentrated among a small set of movies.
- Personalized recommendations are necessary to surface relevant content beyond popular titles.
- Filtering improves computational efficiency but excludes cold-start users and less popular movies.

4. Methodology:

Two recommendation approaches were implemented and compared: **Item-Based Collaborative Filtering (Item-CF)** and **Singular Value Decomposition (SVD)**. Both models were trained and evaluated using the same train-test split.

Train-Test Split
A per-user time-based split was used to simulate a real recommendation scenario. For each user, the earliest 80% of ratings were used for training and the most recent 20% for testing.

Dataset	**Ratings**
Training Set	941,270
Test Set	233,990

A movie was considered relevant if its actual rating was **≥ 3.5**.

4.1 Item-Based Collaborative Filtering

Item-CF assumes that movies receiving similar ratings from users are similar to each other. A sparse user-item matrix was constructed and cosine similarity was computed between movies. Ratings were predicted using a similarity-weighted average of the 30 most similar

movies previously rated by the user. Recommendations were generated by ranking unseen movies according to their predicted ratings.
Advantage: Simple and interpretable.
Limitation: Predicted ratings tend to cluster around the middle of the rating scale.

4.2 Singular Value Decomposition (SVD)

SVD is a matrix factorization technique that learns latent factors representing user preferences and movie characteristics. Ratings are predicted from the interaction of these latent factors together with user and movie biases.

Hyperparameter	Value
Latent Factors	100
Epochs	20
Learning Rate	0.005
Regularization	0.02

SVD was implemented using the Surprise library and trained on the same dataset split as Item-CF.
Advantage: Better captures global patterns and produces well-calibrated predictions.
Limitation: Requires model training and is less interpretable than Item-CF.

5. Evaluation & Experimental Results:

The models were evaluated on the same test set using two metrics:

- **RMSE (Root Mean Squared Error):** Measures rating prediction accuracy. Lower values indicate better performance.
- **MAP@10 (Mean Average Precision @ 10):** Measures recommendation ranking quality. A movie was considered relevant if its actual rating was **≥ 3.5**. Higher values indicate better performance.

Model Comparison

Metric	Item-CF	SVD
RMSE	0.9823	0.8455
MAP@10	0.0632	0.1717
Training Time	~0 s	25.3 s
Prediction Time/User	5.55 ms	7.25 ms
Memory Usage	7.72 MB	27.83 MB

Results and Discussion

SVD achieved the best recommendation quality, outperforming Item-CF on both RMSE and MAP@10. The RMSE improvement indicates more accurate rating predictions, while the substantially higher MAP@10 shows that SVD is more effective at placing relevant movies near the top of recommendation lists.

Item-CF was computationally cheaper, requiring no explicit training phase and slightly less memory on this dataset. However, its recommendation quality was noticeably lower. Overall, SVD provides the best balance between rating accuracy and recommendation quality, while Item-CF serves as a simple and interpretable baseline.

6. Recommendation Examples:

To evaluate recommendation quality in practice, the Top-10 recommendations generated by each model were compared against users' actual test-set favourites. A recommendation was considered a **hit** if it appeared in the Top-10 list and the user rated it **≥ 3.5** in the test set.

Aggregate Hit Analysis

Measure	Item-CF	SVD
Average Top-10 Hits/User	1.44	2.91
Users with ≥ 1 Hit	62.0%	91.5%

SVD achieved nearly twice as many hits per user and successfully recommended at least one relevant movie for over 90% of users, confirming its superior ranking performance.

Success Case (User 57186)

This user preferred highly rated drama and thriller films such as *The Godfather*, *The Deer Hunter*, and *The Silence of the Lambs*. SVD successfully recommended several of the user's future favourites, including *Lord of the Rings: The Fellowship of the Ring*, *Batman Begins*, and *Ray*. In contrast, Item-CF recommended niche arthouse films that did not match the user's interests.

Failure Case (User 188190)

This user exhibited highly diverse preferences spanning multiple genres. SVD generated a coherent set of recommendations, but none matched the user's future favourites. This highlights a limitation of recommendation systems: users with broad and inconsistent tastes are inherently more difficult to model accurately.

Key Observations

- SVD produces more personalized recommendations than Item-CF.
- Item-CF tends to recommend similar niche titles to different users.
- SVD performs well for users with clear preference patterns.
- Both models struggle when user interests span many unrelated genres.

7. Key Insights & Future Work:

Key Insights

- Rating prediction and recommendation ranking are different objectives. While SVD outperformed Item-CF on both metrics, the improvement was much larger for MAP@10 than for RMSE, highlighting the importance of evaluating recommendation quality separately from rating accuracy.

- SVD produced more personalized and better-calibrated recommendations than Item-CF. Its latent-factor approach captured global user preferences more effectively, whereas Item-CF often recommended similar niche titles to different users.
- More data does not always improve recommendation quality. In the scale-up experiment, adding a second Netflix rating file improved RMSE but reduced MAP@10 because the larger movie catalogue made ranking relevant movies more challenging.

Metric	Small Dataset	Large Dataset
RMSE	0.8455	0.7975
MAP@10	0.1717	0.1274
Training Time	25.3 s	114.9 s

- The choice of model depends on application requirements. SVD offers superior recommendation quality, while Item-CF remains useful when interpretability and training-free deployment are important.

Future Work

- Develop cold-start solutions for new users and rarely rated movies using content-based or hybrid approaches.
- Incorporate movie metadata such as genres, release years, and descriptions to improve recommendation quality.
- Evaluate additional ranking metrics such as NDCG, coverage, diversity, and hit rate.
- Scale the system to the complete Netflix Prize dataset and explore alternative matrix factorization methods such as ALS.